
COMPUTER VISION - PROJECT 1

Computer Vision 2025-26

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Topic: Binary classification of skin lesion images

Goal: Extract features from skin lesion images and use them to train a machine learning classifier to predict the lesion class

Scenario

In this project, you will work with dermoscopic images belonging to two distinct classes: benign nevi (moles) and malignant melanoma. Dermoscopic images show characteristic visual patterns that can guide feature design in computer-aided analysis. A widely used clinical framework to describe these differences is the ABCD rule of dermoscopy, which highlights morphological properties commonly associated with melanoma.

1. *Melanocytic Nevus* (Benign Mole)

A nevus is a common, benign proliferation of melanocytes. In dermoscopic images, nevi typically exhibit:

- *Symmetry*: The lesion is often symmetric in shape and pattern.
- *Regular Borders*: Edges are smooth and well-defined.
- *Uniform Color*: One or two colors are usually present (e.g., light brown, dark brown).
- *Homogeneous Structure*: Patterns such as a regular pigment mesh.

These visual regularities mean that benign nevi often produce stable geometric features, uniform color distributions, and consistent texture patterns.

2. *Melanoma* (Malignant Skin Lesion)

Melanoma is a malignant tumour arising from melanocytes. Early recognition is crucial, and dermoscopic images commonly show:

- *Asymmetry*: Uneven shape or structure across one or two axes.
- *Border Irregularity*: Notched, blurred, or highly uneven edges.
- *Color Variegation*: Multiple colors may coexist (brown, black, red, white, blue).
- *Diameter / Dynamics*: Larger size or noticeable evolution over time.

These characteristics correspond to higher spatial variability, irregular edge features, heterogeneous color distributions, and complex texture patterns.

You can select or design feature extraction methods—such as texture descriptors, shape analysis, edge detectors, or color-based metrics—that capture clinically meaningful patterns.

Your Task:

Write a Python program which includes the following steps:

1. Lesion Segmentation

The first step is to isolate the lesion from the surrounding skin. Segmentation reduces background variability—such as skin color, hair, or artifacts—and ensures that subsequent analysis focuses on the lesion itself.

You can use classical computer vision techniques like thresholding, region growing, or clustering (e.g., k-means in color space) to produce effective lesion masks.

After segmentation, apply the mask to extract the lesion region, eliminating background skin and allowing cleaner, more focused analysis.

2. Image Pre-processing

Once the lesion is extraction isolated, the next step is to improve the image quality and reduce noise. Typical pre-processing tasks include:

- *Noise Reduction:* Median filtering, Gaussian smoothing, or bilateral filtering to reduce sensor noise and artifacts.
- *Contrast Enhancement:* histogram equalization can help to highlight important structures.

These operations help standardize your dataset and improve the reliability of features.

3. Feature Extraction

Using the clinical characteristics described earlier (asymmetry, border irregularities, color variation, and diameter), you can extract computational features that describe the lesion. These may include shape features, border features, color features, and texture features. These engineered features form the numerical representation of each lesion used by the classifier.

4. Classification

Finally, use the extracted features to train a machine learning model to classify lesions into the two target categories: melanoma and nevus and evaluate the performance of your classifier. The goal is to determine how well your chosen features and classifier can distinguish between the two lesion types.

Dataset

Data can be downloaded from the following link:

<https://www.kaggle.com/datasets/andrewmvd/isic-2019?resource=download>

The dataset contains images for nine different diagnostic categories but you have to use only melanocytic nevus and melanoma images.

Optional

Beyond the techniques covered in the course, you may also take inspiration from the scientific literature and experiment with more advanced descriptors. This is optional but can enrich your feature set and potentially improve classification performance.

- Try more advanced segmentation schemes, e.g., active contours (snakes) or watershed.
- Use *Deep Learning Approaches* for segmentation: U-Net, DeepLab, and other encoder–decoder architectures are widely used for medical image segmentation. Pre-trained segmentation models available online may be used, as this is a preliminary step and the goal is not to train a segmentation model from scratch.
- *Advanced Contrast Enhancement*: adaptive contrast enhancement, or CLAHE (Contrast Limited Adaptive Histogram Equalization) can be used to highlight important structures.
- Use color normalization or illumination correction techniques for reducing differences caused by lighting conditions or camera variability.